

# Hospital Data Warehouse & BI Project

## Final Project Report

---

### 1. Project Overview

This project focuses on designing and implementing a **Data Warehouse and Business Intelligence solution** for a hospital scenario.

The system integrates data from multiple sources, applies ETL processes for cleaning and transformation, and delivers analytical dashboards using Power BI to support decision-making.

---

### 2. Business Scenario

**Hospital Analytics** was selected to analyze:

- Patient encounters
- Healthcare costs
- Insurance coverage
- Hospital and payer performance
- Trends over time

This scenario is well-suited for dimensional modeling and analytical reporting.

---

### 3. Data Sources

To meet project requirements, two heterogeneous data sources were used:

- **CSV Files:**  
patients, encounters, procedures
- **SQL Server Database:**  
organizations (hospitals), payers (insurance companies)

This fulfills the requirement of using different data formats.

---

## 4. Database Architecture

A dedicated Data Warehouse database named **Hospital\_DW** was created and organized using schemas:

- **stg**: Staging layer for raw and cleaned data
- **dw**: Data warehouse layer for analytics

This separation improves clarity and maintainability.

---

## 5. ETL & Staging Process

The ETL process was implemented using SQL and divided into two stages:

- **Raw Staging:**  
Data loaded “as-is” using NVARCHAR(MAX) to avoid format issues.
- **Clean Staging:**  
Data was transformed using TRY\_CAST, cleaned, and standardized before loading into the data warehouse.

Invalid records were identified and removed to ensure data quality.

---

## 6. Data Warehouse Design

A **Star Schema** was implemented with:

### Dimension Tables:

- DimPatient
- DimOrganization
- DimPayer
- DimDate

### Fact Table:

- FactEncounters

The fact table stores cost measures and links to dimensions using surrogate keys.

---

## 7. Data Quality Validation

Before analysis, data quality checks were performed:

- Detection of missing values
- Validation of cost fields
- Removal of invalid cost records

This ensured reliable and accurate analysis.

---

## 8. Power BI Dashboards

Power BI was connected directly to the SQL Server Data Warehouse.

**Dashboards include:**

- KPIs (total encounters, total cost, average cost, coverage)
- Charts showing trends over time
- Cost analysis by hospital, payer, and gender
- Filters for year, city, encounter class, and payer
- A detailed table showing encounter-level information

These dashboards provide both high-level insights and detailed analysis.

---

## 9. Actionable Insights

The solution helps decision-makers:

- Identify major cost drivers
  - Monitor healthcare trends
  - Compare insurance coverage with actual costs
  - Evaluate hospital and payer performance
- 

## 10. Conclusion

This project demonstrates the full Data Warehouse lifecycle:

- Data integration from multiple sources
- ETL and data cleansing
- Dimensional modeling
- Business Intelligence reporting

The final system delivers meaningful insights and supports data-driven healthcare decisions.

---

### **Final Statement**

The project emphasizes correct data warehouse design and analytical thinking, focusing on understanding concepts rather than relying on complex tools.